Recipe 3

Understand Sequence Alignments

To assess an alignment we need to describe its attributes with proper "terms" and "quantities"

"Verbalizing" alignments

It is a surprisingly challenging task.

You can find substantial training materials spent on explaining how the aligner works internally:

> "...a dynamic progamming algorightm with diagonalization and memory optimization ..." ”

At the same time you will find suprisingly sparse information on **what** gets reported and **how** it gets reported.

Alignment attributes

1. Alignment length: how many bases of the "top" sequence are covered by the "bottom" sequence
2. Percent identity: what percent of the top sequence exactly macthes in the bottom sequence
3. Mismatches: how many mismatching bases are lined up in the alignmment?
4. How many deletions? How long is the deletion?
5. How many insertions? How long is each insertion? What has been inserted?

... many more terms

More subtle details

1. Will the aligner report more than one alignment?
2. Will the aligner report all alignments within a certain range?
3. Will the aligner at least mention that there may be other similarl matches?
4. Will the aligner report the variation itself, or just report that there are variations?

... and so on ...

Aligments are information

dense "concepts"

Software may be equal from "algorithm"

point of view but not equal in their utility of what they report and how they report it

Recipe Purpose

In this recipe we demonstrate the output of several tools.

Your job is to:

- Investigate each alignment output.
- Analyze the differences between the outputs.
- See if you can answer questions you might have about an alignment

Recipe Code

The recipe aligns two strains of the Ebola genome

- **1972 (Mayinga)**
- **2014 (Makondo)**

It aims to provide you with an understanding of how the 1972 strain is different from the 2014 strain at sequence level.

The **same** two sequences will be aligned with **different** tools.

Each tool will be run at least twice to produce format their output differently.

Recipe outputs

1. Global Alignment

Runs the `strecher` tool three times, each run customized to format the **alignment** differently.

2. Local Alignment

Runs `blastn` twice for two different outputs.

3. Semi Global Alignment

Runs `minimap2` twice with two different output formats.

See video for discussion on each file.

When would you use each tool?

Visually checking an aligment: `strecher`

Searching for local alignments: `blastn`

Aligning many long sequences against reference genomes: `minimap2`

The following section is optional

It was presented in class and I think is quite useful

Alignments can be deceptive

Manually align:

```
GATTACA
```

to

```
GATCA
```

On a piece of paper write them under each other in a way that you think they align best.

Run the alignment with a software

Here is an example:

```
global-align.sh GATTACA GATCA -data EDNAFULL
```

does the output match your alignment?

Radically different results

```
global-align.sh GATTACA GATCA -data EDNAFULL

GATTACA
|||.|
GATCA--
```

wheras a different gap opening penalty produces:

```
global-align.sh GATTACA GATCA -data EDNAFULL -gapopen 8

GATTACA
|||  ||
GAT--CA
```

Wait there is more

We can also do a

```
local-align.sh GATTACA GATCA -data EDNAFULL
                                        -gapopen 1 -gapext 10
```

an that will produce .. analyze this image a bit

```
GATTACA
||  | ||
GA-T-CA
```

See how you can get any alignment you want if you tune the parameters.